



HOUDAIFA DRAHM

Backend & DevOps Engineer

CONTACT

-  houdaifadrahm@gmail.com
-  +212 644 645 877
-  Benguerir UM6P, Morocco
-  houdaifa.dev
-  github.com/Houdaifa1
-  linkedin.com/in/houdaifa-drahm

SKILLS

Languages

- C
- C++
- TypeScript
- JavaScript
- Python
- Bash

Backend

- Node.js
- NestJS
- REST APIs
- WebSockets
- JWT
- Socket.io
- HTTP/1.1
- TCP/UDP

DevOps

- Docker
- Kubernetes
- GitHub Actions
- Nginx
- DigitalOcean
- Linux Admin

Systems

- POSIX
- pthreads
- Mutexes
- Semaphores
- epoll
- poll()
- Signals

Tools & DB

- Git
- Vim
- Valgrind
- GDB
- PostgreSQL
- MariaDB

LANGUAGES

Arabic

Native ●●●●●●

English

Professional ●●●●●●

French

Professional ●●●●●●

Backend & DevOps

Engineer

PROFILE

Backend and DevOps engineer specializing in high-performance Node.js/NestJS microservices and scalable containerized infrastructure. Expert at bridging low-level systems programming (C/C++, POSIX) with modern DevOps automation including CI/CD pipelines, Docker orchestration, and real-time WebSocket architectures. Currently completing the 42 Network engineering curriculum at 1337 School (UM6P).

PROJECT EXPERIENCE

ft_transcendence Full-Stack Real-Time Multiplayer Platform

- Node.js
- NestJS
- Socket.io
- JWT Auth
- GitHub Actions
- DigitalOcean

Full-stack real-time multiplayer platform with WebSocket game rooms, live chat, and JWT refresh-token auth. Zero-downtime deployments via GitHub Actions CI/CD to DigitalOcean.

Webserv Production-Grade HTTP/1.1 Engine in C++

- C++
- poll()/select()
- CGI
- Chunked Transfer
- Zero external libs

HTTP/1.1 server built from scratch in C++ with no external libraries. Non-blocking I/O via poll()/select() with full GET, POST, DELETE, chunked transfer encoding, and CGI support.

Inception Multi-Container Infrastructure Stack

- Docker Compose
- Nginx TLS
- MariaDB
- WordPress
- Custom Networking

Multi-container infrastructure via Docker Compose: Nginx with TLS termination, MariaDB with persistent volumes, and WordPress in isolated containers with custom networking.

cub3D 3D Raycasting Engine and Renderer

- C
- Raycasting
- miniLibX
- Texture Mapping
- Graphics

3D renderer built from scratch in C inspired by Wolfenstein 3D. Real-time raycasting with texture mapping, sprite rendering, and smooth player movement using miniLibX.

Minishell Unix Shell Interpreter

- C
- POSIX
- Pipes
- Signals
- Heredocs
- Valgrind-clean

Fully functional Unix shell in C with Bash-compliant parsing. Supports multi-pipe chaining, heredocs, I/O redirections, environment variable expansion, and signal handling. Valgrind-clean.

Philosophers Concurrent Systems Simulation

- C
- pthreads
- Mutexes
- Deadlock-free
- Stress-tested

Dining philosophers solved with pthreads and mutexes. Guaranteed deadlock-free and starvation-free execution with zero data races under stress testing.

EDUCATION

1337 School (42 Network) — UM6P, Benguerir

2023 – Present

Common Core Program in Systems & Advanced Programming

Project-based peer-reviewed engineering curriculum covering systems programming, networking, OOP in C++, web development, DevOps, and algorithms.

ISTA, Beni Mellal

2021 – 2023

Diploma in Digital Infrastructure, Networking & System Administration

Computer networking, IP addressing, Linux system administration, server configuration, virtualization, and network security.